Experiment No:-2

Title:-

Study on selection of Sensors such as IR sensors, Proximity Sensor, UltrasonicSensor, White line sensor, Temperature Sensor, Touch sensor, Tilt Sensor ,Accelerometer, Gyroscopic Sensor etc. based on given application.

1.A conveyor system must separate metallic objects from plastic components automatically.

- Which sensor will you select?



Fig 1

- Why is it suitable for this application?

1.Detects only metallic objects using electromagnetic induction.

2.Ignores plastic components automatically.

3.Fast response and reliable for conveyor automation.

4.No physical contact required.

2. A storage tank contains chemically reactive liquid. The level must be measured without physical contact.

- Select the appropriate sensor



Fig 2

- Justify your choice based on safety and reliability.

1.Non-Contact Measurement :-

- The sensor is mounted at the top of the tank.
- Uses ultrasonic sound waves to measure liquid level.
- No direct contact with the chemically reactive liquid.

2. High Safety :-

- Prevents corrosion or chemical damage to sensor parts.
- Suitable for hazardous, toxic, or reactive liquids.
- No risk of contamination.

3.Reliable Operation :-

- Works independently of liquid color or transparency.
- Provides continuous level monitoring.
- Minimal maintenance required.

3. A robot must detect obstacles within a range of 20 cm to 2 meters.

- Choose between IR and ultrasonic sensor.



Fig 3

- Justify your selection with technical reasoning.

Selected Sensor: Ultrasonic Sensor

- Technical Justification:

1.Accurate in the 20 cm–2 m range.

2.Works on sound waves, not affected much by object color.

3.Better than IR for varying surface types and lighting conditions.

4. A robot must follow a black line on a white floor in an industrial warehouse.

• Which sensor is suitable?



Fig 4

- Explain how surface contrast affects detection.

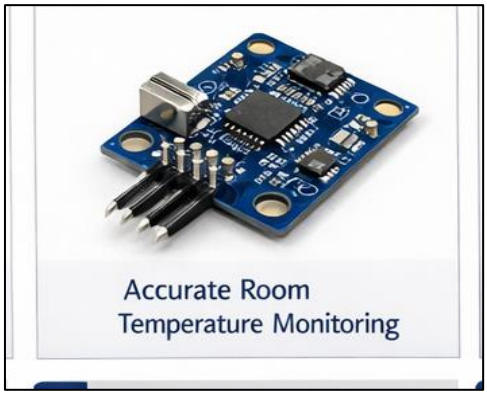
1.White surface reflects IR light → High signal.

2.Black line absorbs IR light → Low signal.

3.Sensor detects this contrast difference to follow the path.

5. A room automation system must continuously monitor temperature for HVAC control.

- Select suitable temperature sensor.



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Fig 5

- Why would a thermocouple not be preferred for normal room conditions?

Selected Sensor: Digital Temperature Sensor (e.g., LM35 / DHT22)

- Why not Thermocouple?

1.Thermocouples are designed for very high temperatures.

2.Lower accuracy at room temperature.

3.Requires additional signal conditioning.

4.Costlier and unnecessary for normal HVAC range (20–40°C).

6. A quadcopter requires stable flight and tilt correction.

- Which sensors are necessary?



Fig 6

- Why is sensor fusion important?

Required Sensors:

- Accelerometer
- Gyroscope
- (Combined as IMU – Inertial Measurement Unit)

Why Sensor Fusion?

- Accelerometer measures tilt (gravity reference).
- Gyroscope measures angular rotation.

- Fusion improves accuracy, reduces drift, ensures stable flight.

7. A rotating motor must be monitored for abnormal vibration to prevent failure.

- Which sensor is most appropriate?



Fig 7

Selected Sensor: Vibration Sensor

- What parameter does it measure?

Measured Parameter:

1.Vibration amplitude

2.Acceleration (m/s^2)

3.Frequency changes

Used for predictive maintenance and fault detection.

8. A commercial building requires automatic door opening when a person approaches.

- Select suitable sensor**.**



Fig 8

- Mention environmental factors affecting performance.

Selected Sensor: PIR Motion Sensor

Environmental Factors Affecting Performance:

1.Temperature variations

2.Direct sunlight

3.Air drafts

4.Presence of heat sources

Detects human body infrared radiation.

9. A mobile device must detect orientation changes (portrait/landscape).

- Which sensor is used?



Fig 9

- Explain briefly how it detects motion.

Selected Sensor: Accelerometer

How it detects motion:

1.Measures acceleration along X, Y, Z axes.

2.Detects gravity direction.

3Determines portrait or landscape orientation.

10. A car requires rear obstacle distance measurement up to 3 meters.

- Which sensor will you choose?



Fig 10

- Mention limitations under rain or extreme weather.

Selected Sensor: Ultrasonic Parking Sensor

Limitations in Rain/Extreme Weather**:**

1.Heavy rain may scatter sound waves.

2.Mud/dirt on sensor reduces accuracy.

3.Extreme temperature affects sound speed slightly